

Fibonacci Heap

Outperforming Traditional Priority Queues

Pedro Marinho Miguel Mateus Tuomas Haapasalo

University of Porto FEUP

March 23, 2026

Overview

Motivation

Data Structure Basis

Operations

Applications

Example

Conclusions

References

Overview

Motivation

Data Structure Basis

Operations

Applications

Example

Conclusions

References

Motivation

- ▶ Efficient priority queues are a key problem in the context of graph algorithm design.
- ▶ There is an increased need for processing massive networks, but the repeated cost of updating node priorities poses a huge bottleneck!
- ▶ Demand for priority queue designs capable of extremely fast updates, but that can simultaneously keep other standard operations efficient.

The $O(\log n)$ Bound

Currently, the effects of logarithmic slowness are much more visible! The number of edges in dense networks vastly exceeds the vertices, making priority updates the dominant computing cost [1].

Overview

Motivation

Data Structure Basis

Operations

Applications

Example

Conclusions

References

Data Structure Basis - Slide 1

Data Structure Basis - Slide 2

Data Structure Basis - Slide 3

Overview

Motivation

Data Structure Basis

Operations

Applications

Example

Conclusions

References

Operations - Slide 1

Operations - Slide 2

Operations - Slide 3

Operations - Slide 4

Operations - Slide 5

Overview

Motivation

Data Structure Basis

Operations

Applications

Example

Conclusions

References

Applications - Slide 1

Applications - Slide 2

Applications - Slide 3

Applications - Slide 4

Overview

Motivation

Data Structure Basis

Operations

Applications

Example

Conclusions

References

Example - Slide 1

Example - Slide 2

Example - Slide 3

Example - Slide 4

Overview

Motivation

Data Structure Basis

Operations

Applications

Example

Conclusions

References

Conclusions - Slide 1

Conclusions - Slide 2

Overview

Motivation

Data Structure Basis

Operations

Applications

Example

Conclusions

References

References



Michael L. Fredman and Robert Endre Tarjan.

Fibonacci heaps and their uses in improved network optimization algorithms.

Journal of the ACM, 34(3):596–615, 1987.



Thomas H. Cormen, Charles E. Leiserson, Ronald L. Rivest, and Clifford Stein.

Introduction to Algorithms, Third Edition.

MIT Press, 2009.